Project 2: Mile High Stays

Market research for a new pricing strategy

The client

Mile High Stays is a Denver-based vacation-rental management company that has recently been acquired. Its new owners want to update how the company prices its properties. They want prices to be competitive to help increase demand, while reflecting the characteristics of each property.

Business objective

Following its acquisition, Mile High Stays wants to update its Denver pricing model and strategy. Develop a model of advertised nightly prices, use it to recommend initial prices for client-provided properties, and assess whether relationships observed in Seattle and Portland strengthen or qualify your recommendations for Denver.

The client requests

1. Develop a Denver pricing model. Build a linear regression model that describes associations between host and listing characteristics and advertised nightly prices. Use the findings to inform the company's new pricing strategy.

2. Recommend prices for the client's properties. Use your model to predict nightly prices for the 20 properties in `client_properties.csv`. The client requests a table with each listing's identifier, predicted nightly price, and recommended initial nightly price. Briefly explain any differences between predicted and recommended prices. Recommend a pricing strategy and justify it in light of the company's situation and objectives.

3. Compare markets. Estimate models for Seattle and Portland using the same explanatory (X) variables as your Denver model. Compare how amenities and other host or listing characteristics are associated with nightly prices across the three markets. Explain how these comparisons inform your Denver recommendations.

Working with the data

Use the supplied files and the dictionary on the following pages. Choose and justify your variables, data preparation, modeling decisions, and supporting analyses using the material covered through Lecture 11. Document your work so it can be reproduced from the supplied files.

Data dictionary

Market, property, and price variables

`airbnb.csv` contains **6,522 market listings and 19 variables**: 1,717 in Denver, 3,120 in Seattle, and 1,685 in Portland. `client_properties.csv` contains **20 client properties in Denver** with the same characteristics but no price column. Each row represents one listing.

Variable	Definition
`listing_id`	Unique project identifier for a listing. Use it to identify client properties in your recommendations.
`market`	City containing the listing: Denver, Seattle, or Portland.
`price`	Advertised nightly price in US dollars at collection. Quotes may refer to different stay dates. This is not realized booking revenue. Omitted from the client file.
`room_type`	Entire home or Private room. A private room may share other spaces with the host or other guests.
`property_type`	Detailed property category reported in the listing, such as Entire home, Entire rental unit, or Entire townhouse.
`neighborhood`	Neighborhood containing the listing. Neighborhood names are specific to each market.
`accommodates`	Maximum number of guests the listing accommodates.
`bedrooms`	Number of bedrooms. Zero can describe a studio; a blank means unknown.
`bathrooms`	Number of bathrooms parsed from the listing description. Half bathrooms count as 0.5. Counts can include shared bathrooms.
`minimum_nights`	Advertised minimum number of nights required for a stay. Values range from 1 to 7 in this extract.

Data dictionary

Host information, guest feedback, and amenities

Variable	Definition
`superhost`	Whether the host was designated a Superhost: Yes or No.
`number_of_reviews`	Total number of guest reviews recorded for the listing. Zero means no reviews. This is not a count of bookings or guests.
`review_score`	Overall guest review rating, on a scale up to 5. A blank means no rating was recorded.
`cleanliness_rating`	Guest rating of cleanliness, on a scale up to 5. A blank means no cleanliness rating was recorded.
`pets_allowed`	Yes if the amenities list advertises Pets allowed; No if it does not.
`free_parking_on_premises`	Yes if the amenities list advertises free parking on the property, including a driveway or garage; No otherwise.
`free_street_parking`	Yes if the amenities list advertises Free street parking; No otherwise. This does not guarantee an available or reserved space.
`air_conditioning`	Yes if the amenities list advertises air conditioning, including central or portable units; No otherwise.
`amenities`	Full list of advertised amenities, stored as text in JSON list format. A blank means unknown; [] means an empty advertised list.

Reading the values. Blank cells represent unknown values. Amenity indicators describe what is advertised: No means an item is not listed, not that its absence was verified. A missing amenity list produces blank indicators. A listing can advertise both on-premises and street parking.

Scope and source. The market data covers entire homes and private rooms with advertised minimum stays of seven nights or fewer, with one selected listing per host within each market. Client properties and their hosts are excluded from the market file. Inside Airbnb collected the source records in September 2026. See SOURCES.md for collection dates and preparation details.

Deliverables

Technical analysis and managerial recommendation

Deliverable 1: Analysis document

Submit a rendered Word or PDF document containing all code used for data wrangling, analysis, modeling, and visualizations. Include the resulting output and fully rendered plots with descriptive titles and labeled axes. This document is intended for technically focused colleagues at your company.

Use clearly labeled sections, well-commented code, and written descriptions to explain what you did, why you did it, and what the results mean. Someone unfamiliar with your code or data should be able to understand the work. Address all three client requests and include the client-property pricing table.

Deliverable 2: One-page executive summary

Submit a well-written, one-page executive summary for Mile High Stays' new owners. Use short sections with descriptive headings and plain language. The summary should stand on its own and cover:

Business objective. State the problem, why it matters to Mile High Stays, and the business decision your analysis supports.

Approach. Briefly explain at a managerial level what listing data you examined and how your analysis addressed the client's requests.

Key findings and implications. Present the most important results, supported by a few specific numbers or comparisons. Explain what they mean for the company and any limitations that materially affect the decision.

Recommendation. Recommend a concrete pricing strategy tied to your findings and the client's objectives. Explain how the evidence supports your recommendation.

Leave code and detailed technical procedures out of the executive summary. The full property-by-property pricing table belongs in the analysis document; summarize its main implications for management.

Assessment

The separate Rubric V2.pdf describes the seven assessment criteria and the 100-point scoring system. A separate Quarto source file is welcome but is not required; the rendered analysis document must contain the code needed to reproduce your findings.